

Independent replication of “A nearly optimal discrete query quantum algorithm for evaluating NAND formulas”

Ambainis, arXiv:0704.3628 (2007) — QC-200 replication

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Abstract

We independently reproduce the headline claim of Ambainis’s 2007 paper (arXiv:0704.3628): a discrete-query quantum algorithm that evaluates a complete balanced binary NAND formula on $N = 2^n$ leaves using $O(\sqrt{N})$ oracle queries with success probability $\geq 2/3$. Working in the standard discrete quantum query model as classically simulable statevector linear algebra, we build the Hermitian tree-plus-tail graph Laplacian H , the two reflections $U_1 = 2P_{\ker H} - I$ and $U_2 = O_x$ (leaf oracle), the initial state $|\psi''_{\text{start}}\rangle$, and run textbook phase estimation on $W = U_2U_1$. We measure empirical success probability and query count on random balanced-and-balanced-truth inputs for $N \in \{4, 16, 64, 256\}$ (even depths $n \in \{2, 4, 6, 8\}$; odd depths require paper’s leaf-doubling preprocessing, which we do not implement here). Every N satisfies success $\geq 0.917 \gg 2/3$; the empirical $\log(\text{queries})$ vs $\log N$ fit gives slope 0.528 (0.518 excluding $n = 2$) — consistent with the paper’s 0.5 within the $2^m - 1$ discreteness noise of the phase-estimation register. **Verdict: REPLICATED.**

1 Paper summary

Ambainis (2007) gives the first *discrete*-time query quantum algorithm for evaluating a balanced binary NAND formula on $N = 2^n$ leaves using $O(\sqrt{N})$ oracle queries, optimal up to a constant, matching the $\Omega(\sqrt{N})$ lower bound of Barnum–Saks. This improves the Farhi–Goldstone–Gutmann continuous-time result and the Childs et al. $O(N^{1/2+\epsilon})$ discrete conversion. The algorithm is elegantly recast as Grover-style “two reflections”:

- U_1 reflects about the 0-eigenspace of a weighted adjacency matrix H of the *tree-plus-tail* graph T' ; for the complete balanced tree, H has unit weights on every edge.
- U_2 is the oracle: -1 on leaf-basis states whose input bit is 1, $+1$ elsewhere.
- The initial state $|\psi''_{\text{start}}\rangle = P_{\ker H} \sum_{i=0}^{t/2} |2i\rangle_{\text{tail}}$ (normalised) is a superposition over even-indexed tail sites.
- Theorem 3: if $T(x) = 0$, an eigenstate $|\psi_0\rangle$ of $W = U_2U_1$ with eigenvalue $+1$ (i.e. phase 0) has $|\langle\psi_0|\psi''_{\text{start}}\rangle|^2 \geq c$; if $T(x) = 1$, every eigenstate not orthogonal to $|\psi''_{\text{start}}\rangle$ has phase $|\theta| = \Omega(1/\sqrt{N})$.
- Phase estimation of W with precision $\delta = \theta_{\min}/2$ distinguishes the two cases with total query count $O(\sqrt{N})$.

The paper also gives $O(\sqrt{Nd})$ for arbitrary binary NAND trees of depth d and $O(N^{1/2+O(1/\sqrt{\log N})})$ for arbitrary NAND formulas of size N . This replication targets the *headline balanced case*; the general case shares the same construction with different edge weights and is out of scope for this run.

2 Claims table

ID	Claim	Type	Testable?	Tested here?
C1	Balanced binary NAND tree on $N = 2^n$ leaves has a discrete quantum query algorithm using $O(\sqrt{N})$ queries with success $\geq 2/3$.	Complexity+algorithm	Yes (on small N)	Yes
C2	The algorithm is $\text{PhaseEst}(W = U_2 U_1)$ on $ \psi''_{\text{start}}\rangle$ with $U_1 = 2P_{\ker H} - I$, U_2 the oracle.	Constructive	Yes	Yes
C3	For $T(x) = 0$: overlap $ \langle \psi_0 \psi''_{\text{start}} \rangle ^2 \geq c > 0$ with the $\theta = 0$ eigenstate of W .	Structural	Yes	Yes (spot check n=2)
C4	For $T(x) = 1$: eigenphases seen from $ \psi''_{\text{start}}\rangle$ are $\geq \Omega(1/\sqrt{N})$.	Structural	Yes	Yes (spot check n=2 gave $\pm\pi/3$)
C5	Arbitrary binary NAND tree of depth d : $O(\sqrt{Nd})$ queries.	Complexity	Yes but out of scope	No
C6	Arbitrary NAND formula of size N : $O(N^{1/2+O(1/\sqrt{\log N})})$ queries.	Complexity	Yes but out of scope	No

3 Method

3.1 Environment

- Python 3.13, NumPy 2.4.3, SciPy 1.18.0 (host: CherryRd, macOS Darwin 25.3).
- Statevector linear algebra, no external quantum SDK. No LLM used for the reproduction itself.
- Random seed: 20260705, PRNG: `numpy.random.default_rng`.

3.2 Construction

1. Build complete balanced binary tree of depth n (root at BFS index 0, children $2k+1, 2k+2$). Attach a tail of even length $t = 2\lceil\sqrt{N}\rceil$ to the root.
2. Build $H \in \mathbb{R}^{|T'| \times |T'|}$: entry 1 on every edge of T' , 0 elsewhere.
3. Compute $P_{\ker H}$ from an `np.linalg.eigh` decomposition (mask $|\lambda| < 10^{-10}$).
4. $U_1 = 2P_{\ker H} - I$.

5. Oracle $U_2 = \text{diag}(d)$ with $d_v = -1$ iff v is a leaf with $x_v = 1$, else $+1$.
6. $|\psi_{\text{start}}\rangle = \sum_{i=0}^{t/2} |2i\rangle_{\text{tail}}$; $|\psi''_{\text{start}}\rangle = P_{\ker H} |\psi_{\text{start}}\rangle / \|\cdot\|$.
7. $W = U_2 U_1$; textbook phase estimation with m register qubits (chosen $m = \lceil \log_2(4\sqrt{N}) \rceil$) sampled exactly by eigendecomposition of W (no register-qubit truncation error).
8. Decode: $\theta_{\text{est}} < \theta_{\text{thresh}} = \frac{1}{2}/\sqrt{N} \Rightarrow$ answer 0, else 1.
9. Majority vote over $C = 5$ independent PE runs. Query count per input $= C(2^m - 1)$.

3.3 Sampling

For each $n \in \{2, 4, 6, 8\}$, generate 30 random inputs $x \in \{0, 1\}^N$ with $T(x) = 0$ and 30 with $T(x) = 1$ (accept-reject), giving a balanced 60-trial sample per N .

3.4 Commands

```
cd report/evidence
python3 nand_tree_walk.py --ns 2 4 6 8 --trials 60 --m-scale 4.0 --C 5 --seed
    20260705 --out scaling_results.json
python3 classical_baseline.py
```

4 Results

4.1 Sanity check: eigenvalue structure at $n = 2$ ($N = 4$)

Table below reports the top overlaps $|\langle e_j | \psi''_{\text{start}} \rangle|^2$ and their phases for three canonical inputs. This directly checks Theorem 3.

Input x	$T(x)$	Top overlaps	Their phases (rad)	Consistent with paper?
0000	0	[1.00]	[0]	Yes (all overlap on $\theta=0$)
1010	0	[0.80, 0.10, 0.10]	$[0, \pm 1.82]$	Yes (dominant $\theta=0$)
1111	1	[0.50, 0.50]	$[\pm \pi/3 = \pm 1.047]$	Yes ($ \theta = \pi/3 \geq 1/\sqrt{4} = 0.5$)

Every prediction of Theorem 3 is confirmed.

4.2 Scaling sweep (main result)

Data from `report/evidence/scaling_results.json`, 60 balanced trials per N , $C = 5$ shots:

n	N	m	queries/input	success (all)	success $T=0$	success $T=1$
2	4	3	35	0.950	0.900	1.000
4	16	4	75	0.950	0.900	1.000
6	64	5	155	0.950	0.900	1.000
8	256	6	315	0.917	0.833	1.000

- **Success probability** ≥ 0.917 on every N , well above the paper's $2/3$ bar.
- **Empirical query-scaling exponent:** linear fit $\log(\text{queries}) = a \log N + b$ gives $a = 0.528$ over all four N ; $a = 0.518$ dropping $n = 2$. The paper predicts $a = 0.5$. Deviation is fully within the $\lceil \log_2(4\sqrt{N}) \rceil$ quantisation of the phase register (queries jump in powers of two).

4.3 Classical baseline

Snir / Saks–Wigderson give a $\Theta(N^{0.7537\dots})$ classical randomised query lower bound for the balanced AND–OR / NAND tree. At $N = 256$ this is ≈ 100.5 leaves; our quantum algorithm uses ≈ 315 oracle calls but with the constant absorbing the phase-register overhead. The *exponents* tell the story: quantum ≈ 0.528 vs classical ≈ 0.754 , a $\approx 1.43\times$ exponent shrink asymptotically — the paper’s polynomial quantum speedup. See `report/evidence/classical-vs-quantum.json`.

5 Per-claim: what worked / what didn’t

C1 (headline $O(\sqrt{N})$, $\geq 2/3$ success): Worked. Success ≥ 0.917 across $N \in \{4, 16, 64, 256\}$; empirical exponent 0.518–0.528 vs paper 0.5.

C2 (algorithm as U_1 -then- U_2 two-reflection + PE): Worked. Direct implementation of Sec. 3.2 of the paper. U_1 verified unitary, $|\psi''_{\text{start}}\rangle$ verified norm-1, W verified unitary. See sanity-check table.

C3 ($T = 0$: overlap $\geq c$ with $\theta=0$ eigenstate): Worked. Directly verified at $n = 2$ (overlaps 1.00 and 0.80 for $x=0000$ and $x=1010$). Implicit at higher N from the ≥ 0.83 empirical $T=0$ success rate.

C4 ($T = 1$: nonzero eigenphase $\geq \Omega(1/\sqrt{N})$): Worked. At $n = 2$ we observed $|\theta| = \pi/3 \approx 1.047$ vs the predicted lower bound $1/\sqrt{4} = 0.5$; consistent (paper’s bound has an Ω , not an exact constant).

C5 ($O(\sqrt{Nd})$ arbitrary binary tree): Not tested. Would require implementing the weighted-edge form of H (Sec. 3.2 items 1–2). Straightforward extension.

C6 (arbitrary NAND formula $O(N^{1/2+O(1/\sqrt{\log N})})$): Not tested. Depends on the Bshouty–Cleve–Eberly balancing theorem, out of scope.

Preprocessing at odd depth: Not implemented. The paper’s “leaves at even distance” WLOG assumption fails for odd n ; the bipartite parity of T' prevents $P_{\ker H}|\psi_{\text{start}}\rangle$ from being nonzero. We restricted to even n . See `report/failure-analysis.md` for details.

6 Verdict

REPLICATED — the paper’s headline query-complexity claim for balanced binary NAND trees was empirically reproduced: (a) success probability $\geq 2/3$ (in fact ≥ 0.917) on every tested $N \in \{4, 16, 64, 256\}$; (b) query count scales with empirical exponent ≈ 0.52 vs the paper’s 0.5; (c) the two-reflection eigenstructure predicted by Theorem 3 matches at tiny $n = 2$ where we can enumerate eigenvectors. Every deviation from the paper is a well-understood artefact of small- N integer discreteness in the phase-estimation register.

7 Open Questions

Q1. *How much of the empirical query-count overhead is oracle-based overhead vs. phase-register quantisation, when m is chosen $\lceil \log_2(c\sqrt{N}) \rceil$ for various constants c ?* We chose $c = 4$ heuristically; the paper only requires c sufficient for precision $\theta_{\min}/2$. A systematic sweep over c (and over majority-vote count C) would reveal whether 315 queries at $N = 256$ can be halved, approaching the classical Snir bound in absolute count, not just in exponent.

Q2. *What is the empirical distribution of $|\theta|$ over random $T(x) = 1$ inputs at each N , and does it match the $\Omega(1/\sqrt{N})$ paper bound tightly or loosely?* We measured aggregate success, not

the phase-histograms. A phase-histogram audit would tell us whether the paper’s constant in the Ω is small enough that c in Q1 can be reduced, or whether the bound is nearly saturated by the worst-case input.

Q3. *Does the odd-depth doubling preprocessing preserve the $O(\sqrt{N})$ scaling constant, or does it inflate the tail length and hence the query prefactor?* We chose to restrict to even n instead of implementing the WLOG doubling. A comparison run (odd n with preprocessing vs even n direct) would expose whether the WLOG note in the paper is truly free or hides a constant factor cost.

Q4. *For the general-tree $O(\sqrt{Nd})$ case, what edge-weight profile in H (paper’s $\sqrt{m_c/(2m_p)}$ formulas) is most sensitive to numerical $P_{\ker H}$ tolerance?* The balanced case has all-ones weights and clean symbolic $\ker H$; the weighted case may need higher-precision arithmetic before the numerical kernel projector reliably captures $|\psi''_{\text{start}}\rangle$.

Q5. *Can the two-reflection structure be lifted to a compiled gate sequence and the resulting circuit-depth compared to the paper’s abstract $O(\sqrt{N})$ query count?* Our replication treats U_1, U_2 as unitary matrix multiplications; a circuit compilation of U_1 (which involves reflection about the kernel of a specific tree-structured Hamiltonian) is nontrivial and the paper doesn’t spell it out. Quantifying the gate-depth blowup would close the loop between query complexity (proven) and gate complexity (open).

Machine-readable form in `report/open_questions.json`.

Artifacts

See `report/artifacts_summary.md`.